

Quantum Computing Mock Final Exam

Part I - Multiple Choice

1. What does a Hadamard gate do?
A. Measures a qubit B. Creates superposition C. Swaps qubits D. Resets qubit
2. Which gate flips $|0\rangle$ to $|1\rangle$?
A. H B. X C. Z D. S
3. What does an Oracle do in Grover's algorithm?
A. Measures qubits B. Flips the phase of the marked state C. Creates entanglement only D. Removes incorrect states
4. What is the purpose of the diffusion operator?
A. Increase marked state's probability amplitude B. Reset qubits C. Measure D. Encrypt
5. A 3-qubit register has how many basis states?
A. 4 B. 6 C. 8 D. 16
6. Measurement causes what?
A. Entanglement B. Collapse of the quantum state C. Superposition D. Teleportation
7. Which company primarily builds trapped-ion quantum computers?
A. IBM B. Google C. IonQ D. Rigetti
8. D-Wave systems primarily use:
A. Superconducting gates B. Quantum annealing C. Photonics D. Neutral atoms
9. Which Qiskit object stores measurement results?
A. QuantumRegister B. ClassicalRegister C. Backend D. Job
10. Why are many shots executed?
A. Faster compile B. Estimate measurement probabilities C. Add qubits D. Reduce gates

Part II - Short Answer

11. Explain superposition.
12. Explain entanglement.
13. Why must measurement occur after the oracle and diffusion operator?
14. Describe the five major steps of Grover's algorithm.
15. What information does a counts histogram provide?

Answer Key

1. B. Creates superposition
2. B. X
3. B
4. A
5. C. 8
6. B
7. C
8. B
9. B

10. B

11. A qubit exists in a linear combination of $|0\rangle$ and $|1\rangle$ until measured.

12. Entangled qubits have correlated states that cannot be described independently.

13. Early measurement collapses the quantum state and destroys the algorithm.

14. Initialize $\rightarrow$ Oracle $\rightarrow$ Diffusion $\rightarrow$ Measure $\rightarrow$ Execute/Analyze.

15. It shows how frequently each measured bitstring occurred over all shots.